

Finite-rank aDP operators attain their norm

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Source and target

The source paper is Kadets, Martín, Merí, and Pérez, *Spear operators between Banach spaces*, arXiv:1701.02977. All Banach spaces below are real or complex. The source paper's Chapter 10 asks, among other things, whether the rank-one results for spear/aDP/lush operators extend to finite-rank operators, and also asks whether every spear operator attains its norm.

This note proves a partial finite-rank norm-attainment result. It does not solve the full finite-rank lushness problem in the original codomain.

Definitions used

A norm-one operator $G : X \rightarrow Y$ has the alternative Daugavet property (aDP) if

$$\|G + \omega T\| = 1 + \|T\|$$

for every rank-one operator $T : X \rightarrow Y$ and some unimodular scalar ω depending on T . A spear operator satisfies the analogous identity for all bounded linear $T : X \rightarrow Y$, so every spear operator has the aDP.

We use three stated facts from the source paper.

1. Remark 3.3(ii): if $G : X \rightarrow Y$ has the aDP and $Z \subset Y$ contains $G(X)$, then $G : X \rightarrow Z$ has the aDP.
2. Proposition 6.5: if the codomain is finite-dimensional, then for a norm-one operator $H : X \rightarrow Z$, lushness, spearness, and the aDP are equivalent.
3. Proposition 7.9: if $H : X \rightarrow Z$ is lush, then

$$B_X = \overline{\text{conv}}\{x \in S_X : \|Hx\| = 1\}.$$

Proof Intuition

The finite-rank assumption lets us replace the original codomain by the actual range $Z = G(X)$. That move is one-way for the aDP: the source paper explicitly allows codomain restriction, while also warning that codomain extension can fail. Once the codomain is the finite-dimensional space Z , the source paper's finite-dimensional codomain theorem upgrades the restricted operator from aDP to lush. Lushness then gives norm-attainment, and because Z carries the norm inherited from Y , the norm-attaining vectors for the restricted operator are exactly the norm-attaining vectors for the original finite-rank operator.

Main result

Theorem 1. *Let X, Y be Banach spaces and let $G : X \rightarrow Y$ be a norm-one finite-rank operator. If G has the alternative Daugavet property, then*

$$B_X = \overline{\text{conv}}\{x \in S_X : \|Gx\| = 1\}.$$

In particular, G attains its norm.

Proof. Let $Z = G(X)$, endowed with the norm inherited from Y , and let $H : X \rightarrow Z$ be the same algebraic operator, $Hx = Gx$. Since G has finite rank, Z is finite-dimensional. Also $\|H\| = \|G\| = 1$.

By Remark 3.3(ii) of the source paper, the aDP of $G : X \rightarrow Y$ passes to the restricted-codomain operator $H : X \rightarrow Z$. Since Z is finite-dimensional, Proposition 6.5 of the source paper says that H is lush. Proposition 7.9 of the same paper therefore yields

$$B_X = \overline{\text{conv}}\{x \in S_X : \|Hx\| = 1\}.$$

Finally, Z has the norm inherited from Y , so $\|Hx\| = \|Gx\|$ for every $x \in X$. This is exactly the displayed conclusion. $\square$

Corollary 1. *Every finite-rank spear operator $G : X \rightarrow Y$ attains its norm. More precisely,*

$$B_X = \overline{\text{conv}}\{x \in S_X : \|Gx\| = \|G\|\}$$

for every nonzero finite-rank spear operator G .

Proof. After normalizing G , spearness implies the aDP, so the theorem applies. Scaling back gives the displayed equality. $\square$

What remains open

The proof deliberately avoids claiming that $G : X \rightarrow Y$ is lush in the original codomain. The source paper states that aDP is preserved when one restricts the codomain to a subspace containing the range, but not under arbitrary codomain extension. Thus the range-restricted operator $H : X \rightarrow G(X)$ is lush, while the original finite-rank operator $G : X \rightarrow Y$ may still resist the rank-one-to-finite-rank lushness problem. The same obstruction prevents the argument from proving the full finite-rank analogue of the source paper’s adjoint-lushness rank-one result.

Verification and novelty notes

The proof is a formal combination of three labeled results in arXiv:1701.02977: Remark 3.3(ii), Proposition 6.5, and Proposition 7.9. The set of norm-attaining vectors is unchanged under the range-codomain restriction because the range carries the inherited norm.

Cheap run-index searches found no prior packet for arXiv:1701.02977 or for this finite-rank aDP norm-attainment consequence. Local checks against the later papers arXiv:2306.02645 and arXiv:2306.02638 did not find this statement. Web searches on 2026-06-16 for exact combinations of “finite-rank”, “alternative Daugavet property”, “spear operators”, and “norm-attaining” found no exact duplicate. Because the proof is a short derived corollary from the source paper, this packet should be reviewed as a modest but useful partial result, not as a full solution of the finite-rank lushness problem.

References

- [1] V. Kadets, M. Martín, J. Merí, and D. Pérez, *Spear operators between Banach spaces*, arXiv:1701.02977.